ARJUNA JEE 2027

Lecture -06 | Mathematics

SEQUENCE & SERIES

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Topics *to be covered*

A Harmonic Progression (H.P.)

B Inserting Means between Two Numbers

C Question Practice

D



Question Practice

Question [JEE Mains-2020]

If $2^{10} + 2^9 \cdot 3^1 + 2^8 \cdot 3^2 + \ldots\ldots + 2 \cdot 3^9 + 3^{10} = S - 2^{11}$, then S is equal to

Ans: D

A $2 \cdot 3^{11}$

B $3^{11} - 2^{12}$

C $\frac{3^{11}}{2} + 2^{10}$

D 3^{11}

Acidic Question

Find the sum of series $\frac{3}{19} + \frac{33}{19^2} + \frac{333}{19^3} + \frac{3333}{19^4} + \ldots\ldots \infty$

$\frac{19}{54}$

$$S = \frac{3}{3}\left\{\frac{9}{19} + \frac{99}{19^2} + \frac{999}{19^3} + \frac{9999}{19^4} + \ldots\ldots\infty\right)$$

(10−1, 100−1, 1000−1, 10000−1)

$$S = \frac{1}{3}\left(\frac{10}{19} - \frac{1}{19} + \frac{100}{19^2} - \frac{1}{19^2} + \frac{1000}{19^3} - \frac{1}{19^3} + \ldots\ldots\infty\right)$$

$$S = \frac{1}{3}\left\{\left(\frac{10}{19} + \left(\frac{10}{19}\right)^2 + \left(\frac{10}{19}\right)^3 + \ldots\ldots\infty\right) - \left(\frac{1}{19} + \frac{1}{19^2} + \frac{1}{19^3} + \ldots\infty\right)\right\}$$

$a = \frac{10}{19}$, $r = \frac{10}{19}$

$a = \frac{1}{19}$, $r = \frac{1}{19}$

IMPORTANT PROPERTIES OF G.P.

(i) If a, b, c are in G.P. $\Rightarrow \frac{b}{a} = \frac{c}{b} \Rightarrow b^2 = ac$ ★★★★★

(ii) 3 numbers in G.P. $\Rightarrow \frac{a}{r}, a, ar$ → $\frac{a}{r}, r$

4 numbers in G.P. $\Rightarrow \frac{a}{r^3}, \frac{a}{r}, ar, ar^3$ → $\frac{a}{r^3}, r^2$

5 numbers in G.P. $\Rightarrow \frac{a}{r^2}, \frac{a}{r}, a, ar, ar^2$ → $\frac{a}{r^2}, r$.

When product of numbers is involved.

IMPORTANT PROPERTIES OF G.P.

(iii) If $(a_1, a_2, a_3, \ldots a_n) \rightarrow$ G.P.

then : $ka_1, ka_2, ka_3, \ldots\ldots\ldots ka_n \rightarrow$ G.P.

$( )^r$ $\quad \frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \ldots\ldots, \frac{1}{a_n} \rightarrow$ G.P.

$a_1^r, a_2^r, a_3^r, \ldots., a_n^r \rightarrow$ G.P.

$a_1 \pm k, a_2 \pm k, a_3 \pm k, \ldots\ldots., a_n \pm k \rightarrow$ not in G.P.

$( )^{\frac{1}{2}}$ $\sqrt{2}, \sqrt{4}, \sqrt{8}, \sqrt{16}, \sqrt{32} \rightarrow$ G.P.

$( )^3$ $2^3, 4^3, 8^3, 16^3, 32^3 \rightarrow$ G.P.

$2, 4, 8, 16, 32 \rightarrow$ G.P.

$\times 3$ $\quad 6, 12, 24, 48, 96 \rightarrow$ G.P.

Recip. $\quad \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{16}, \frac{1}{32} \rightarrow$ G.P.

IMPORTANT PROPERTIES OF G.P.

(iv) If $(a_1, a_2, a_3, \ldots. a_n) \rightarrow$ positive numbers in G.P.

then : $\log_{10} a_1, \log_{10} a_2, \log_{10} a_3, \ldots\ldots, \log_{10} a_n \rightarrow$ A.P.

$a, ar, ar^2, ar^3, \ldots. \rightarrow$ G.P.

Take $\log_b(\ )$

$\log_b a, \log_b(ar), \log_b(ar^2), \log_b(ar^3) \ldots.$

A, A+D, A+2D, A+3D.

$\log_b a, \log_b a + \log_b r, \log_b a + 2\log_b r, \log_b a + 3\log_b r \ldots \rightarrow$ A.P.

IMPORTANT PROPERTIES OF G.P.

(v) If $a_1, a_2, a_3 \ldots\ldots\ldots a_n \rightarrow$ G.P.



Then : $a_n^2 = a_{n-k} \cdot a_{n+k}$

(vi) If $a_1, a_2, a_3, \ldots$ and $b_1, b_2, b_3, \ldots$ are in G.P. then

$a_1b_1, a_2b_2, a_3b_3, \ldots\ldots$ are in G.P. ✓

$\frac{a_1}{b_1}, \frac{a_2}{b_2}, \frac{a_3}{b_3} \ldots\ldots\ldots\ldots$ are in G.P. ✓

[Note : No term of G.P. can be zero.]

(v) # G.P.

$a_1, a_2, a_3, a_4 \ldots\ldots a_{47}$ → G.P. (a, r)

$a_n^2 = a_{n-k} \cdot a_{n+k}$ ✓

$a_n \cdot a_n = a_{2n}$

$a_{11} \cdot a_3 = a_8 \cdot a_6$

$a \cdot r^{10} \cdot a \cdot r^2 = a^2 \cdot r^{12}$

$a \cdot r^7 \cdot a \cdot r^5 = a^2 r^{12}$

$a_{33} \cdot a_{12} = a_{40} \cdot a_5 = a_{20} \cdot a_{25}$ ✓

$a_6 \cdot a_6 = a_9 \cdot a_3$ ✓

Question [IIT-JEE-1982]

The third term of a geometric progression is 4. The product of the first five terms is:

A 4^3

B 4^5

C 4^4

D None of these

$a_3 = 4$

$a_1 a_2 a_3 a_4 a_5 = a_3^2 \cdot a_3^2 \, a_3^1$

$= a_3^5 = (4)^5$

$a_1 \cdot a_5 = a_3 \cdot a_3 = a_4 a_2$

(6) (6)

G.P. $\rightarrow a, r$.

T_3 OR $a_3 = ar^2 = 4$ —①

$a \cdot ar \cdot ar^2 \cdot ar^3 \cdot ar^4 = a^5 r^{10} = (ar^2)^5 = 4^5$. ✓

HARMONIC PROGRESSION (H.P.)

$a_1, a_2, a_3, a_4 \ldots\ldots \rightarrow$ H.P.

Reciprocal

$\frac{1}{a_1}, \frac{1}{a_2}, \frac{1}{a_3}, \frac{1}{a_4} \ldots\ldots\ldots \rightarrow$ are in A.P.

ex:- $\frac{1}{4}, \frac{1}{7}, \frac{1}{10}, \frac{1}{13}$ → H.P. ✓

→ 4, 7, 10, 13 → A.P. ✓

ex:- 3, 4, 6, 12 → H.P. ✓

→ $\frac{1}{3}, \frac{1}{4}, \frac{1}{6}, \frac{1}{12}$ → A.P. ✓

$\frac{1}{4} - \frac{1}{3} = \frac{1}{6} - \frac{1}{4} = \frac{1}{12} - \frac{1}{6}$

$\frac{-1}{12} \quad \frac{-1}{12} \quad \frac{-1}{12}$

A.P.

a

d

$T_n = a + (n-1)d$

$S_n = \frac{n}{2}\left(1st + last\right)$

$1 + 2 + 3$

H.P.

$\frac{1}{a}$ ✗

$\frac{1}{d}$

$\frac{1}{a + (n-1)d}$ ✗

$S_n = \frac{1}{\frac{n}{2}\left(1st + last\right)}$ ✗

✗ $\frac{1}{1} + \frac{1}{2} + \frac{1}{3} = \frac{1}{1+2+3}$

$\frac{11}{6} \neq \frac{1}{6}$

IMPORTANT PROPERTIES OF H.P.

(i) If a, b, c are in H.P. then : $\frac{2}{b} = \frac{1}{a} + \frac{1}{c}$ or $b = \frac{2ac}{a+c}$

(ii) No general formula to find out S_n.

(iii) No term of H.P. can be zero.

(iv) Four integers in H.P. are : 3, 4, 6, 12

(v) Take reciprocal of each term & convert H.P. into A.P., Calculate T_n & again take reciprocal.

$a, b, c \rightarrow$ H.P. $\Rightarrow 2\left(\frac{1}{b}\right) = \frac{1}{a} + \frac{1}{c}$

OR

$\frac{1}{a}, \frac{1}{b}, \frac{1}{c} \rightarrow$ A.P.

$2\left(\frac{1}{b}\right) = \frac{1}{a} + \frac{1}{c}$

$\frac{2}{b} = \frac{a+c}{ac}$

$\frac{b}{2} = \frac{ac}{a+c}$

$b = \frac{2ac}{a+c}$

66%

Question

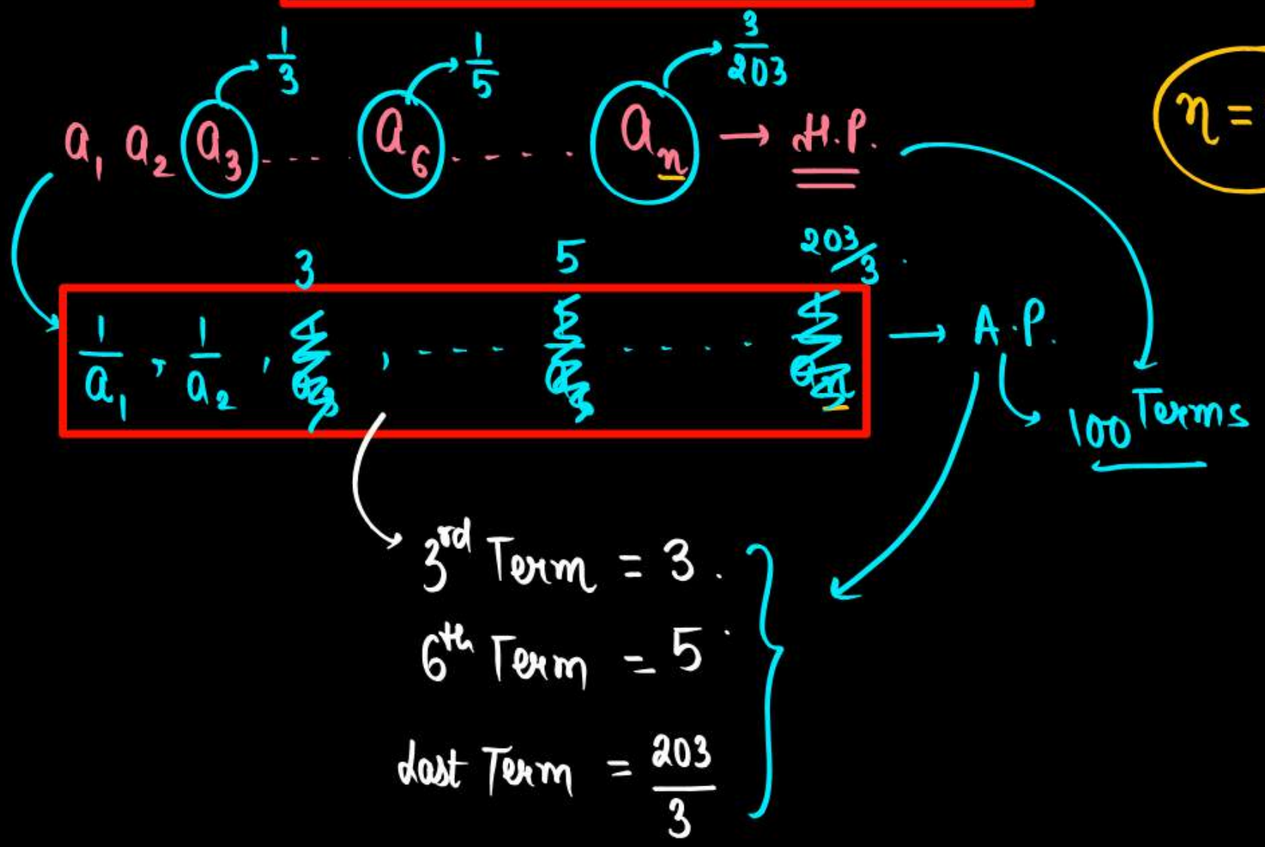
If the 3rd, 6th and last term of a Harmonic Progression (H.P.) are $\frac{1}{3}, \frac{1}{5}$ and $\frac{3}{203}$ respectively, then the total number of terms in the H.P. is.

A 50

B 75

C 100

D 120



AP. (a, d)

3rd Term $= 3 = a + 2d$

6th Term $= 5 = a + 5d$

$\ominus$

$-2 = -3d$

$\frac{2}{3} = d$ ✓

$a = 3 - 2d$

$= 3 - \frac{4}{3}$

$= \frac{5}{3}$ ✓

Last Term $= \frac{203}{3} = a + (n-1)d$

$\frac{203}{3} = \frac{5}{3} + (n-1)\frac{2}{3}$

$203 = 2n + 3$

$200 = 2n$

$100 = n$

Ques:-

If 5th Term is $\frac{4}{5}$

& 11th Term is 11 of a H.P.

Find 13th Term of H.P.

Sol^n:

$\frac{-44}{13}$ → H.P.

A.P. → $\frac{5}{4}$ (5th), ... $\frac{1}{11}$ (11th) ... 13th Term → A.P.

a, d.

13th Term → $\frac{-13}{44}$.

$\Rightarrow a+4d = \frac{5}{4}$

$a+10d = \frac{1}{11}$

$\ominus$

$-6d = \frac{5}{4} - \frac{1}{11} = \frac{55-4}{44} = \frac{51}{44}$

$a = \frac{5}{4} - 4d$, $d = -\frac{51}{4\times 11\times 6_2} = \frac{-17}{88}$

$T_{13} = a + 12d$

$= \frac{5}{4} - 4d + 12d$

$= \frac{5}{4} + 8d$

$= \frac{5}{4} + 8\left(\frac{-17}{88_{11}}\right) = \frac{55-68}{44} = \frac{-13}{44}$

$\therefore f_{nf} = \frac{-44}{13}$

Question

If the pth, q^{th} and r^{th} terms of a Harmonic Progression (H.P.) are a, b and c respectively, then the value of $\frac{q-r}{a} + \frac{r-p}{b} + \frac{p-q}{c}$ is equal to.



A 1

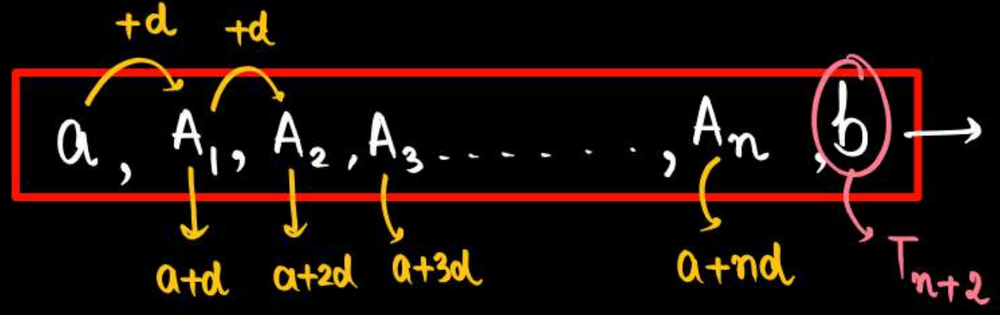
B $p + q + r$

C 0

D abc

INSERTING 'n' A.M.'s

Inserting 'n' A.M.'s between 'a' & 'b' :



first = a.

last = b

Total terms = n + 2.

$T_{n+2} = b$

$a + (n+1)d = b$

$d = \frac{b-a}{n+1}$

Sum of all AM's $= A_1 + A_2 + A_3 + \ldots + A_n$

$= \underline{a+d} + \underline{a+2d} + \underline{a+3d} + \ldots + \underline{a+nd}$

$= na + d(1+2+3+\ldots+n)$

$= na + d\left(\frac{n}{2}(1+n)\right) = na + \frac{n(b-a)}{2}\left(\frac{b-a}{n+1}\right) = na + \frac{nb}{2} - \frac{na}{2}$

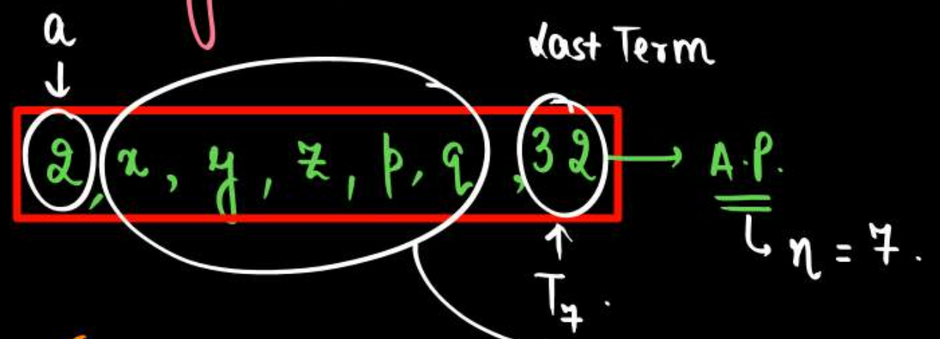
$= \frac{na}{2} + \frac{nb}{2} = n\left(\frac{a+b}{2}\right)$

MOST IMPORTANT OF AM's

Sum of n–AM's inserted between 'a' & 'b' = $n\left(\frac{a+b}{2}\right)$

Ques:- Insert (5) numbers b/w 2 & 32 such that resulting sequence is an A.P.

Soln:

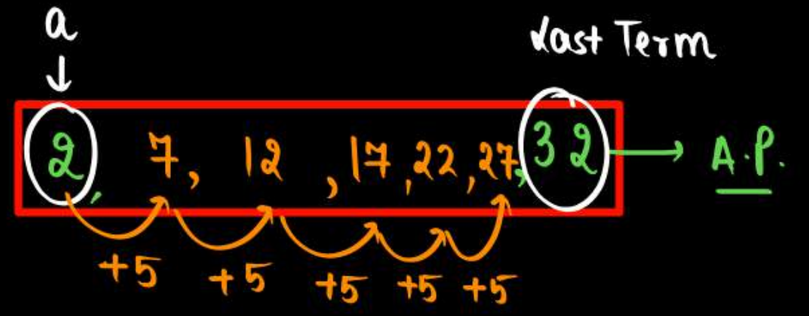


$\therefore n = 7$.

$T_7 = a + 6d = 32$

$2 + 6d = 32$

$d = \frac{30}{6} = 5$.



A_1 A_2 A_3 A_4 A_5

2, 7, 12, 17, 22, 27, 32 → A.P.

5 A.M's inserted b/w 2 & 32.

Sum of 5 AM's $= \frac{5}{2}(2+32)$

$= 5\left(\frac{34}{2}\right)$

$= 5 \times 17 = 85$

Ques:- Insert 6 numbers b/w 2 & 32 such that resulting sequence is an A.P.

8 Terms

$2, A_1, A_2, A_3 \ldots A_6, 32 \rightarrow$ A.P.

Ques:- Insert 15 numbers b/w 2 & 32 such that resulting sequence is an A.P.

17 Terms.

$2, A_1, A_2, A_3 \ldots A_{15}, 32 \rightarrow$ A.P.

Ques:- Sum of (13) AM's inserted b/w 7 & 9.

$7, A_1, A_2 \ldots A_{13}, 9 \rightarrow$ AP

$\text{Sum} = \frac{13}{2}(7+9) = 13 \times 8 = 104.$

INSERTING 'n' G.M.'s

Inserting 'n' G.M.'s between 'a' & 'b' :

$a, G_1, G_2, G_3 \ldots\ldots G_n, b \rightarrow$ G.P.

$ar \quad ar^2 \quad ar^3 \ldots. \quad ar^n \quad T_{n+2}$

first = a

last = b

Total terms = $n+2$

$T_{n+2} = ar^{n+1} = b$

Product of all n-GM's $= G_1 \cdot G_2 \cdot G_3 \ldots\ldots G_n$

$= ar \cdot ar^2 \cdot ar^3 \ldots\ldots ar^n = a^n \cdot r^{1+2+3+\ldots+n}$

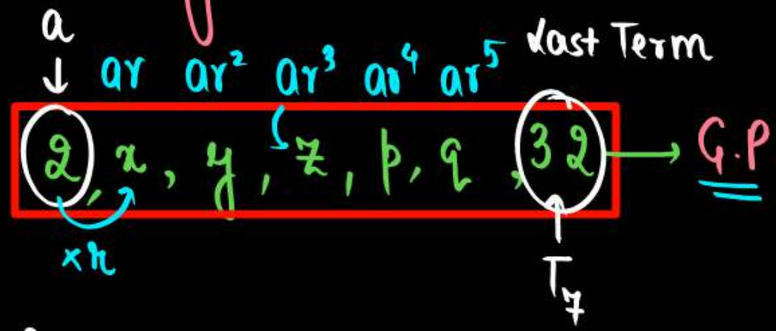
$= a^n \cdot \left(\left(\frac{b}{a}\right)^{\frac{1}{n+1}}\right)^{\frac{n}{2}(1+n)} = a^n \cdot \frac{b^{n/2}}{a^{n/2}} = a^{\frac{n}{2}} b^{\frac{n}{2}}$

$r^{n+1} = \left(\frac{b}{a}\right)$

$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}$

Ques:- Insert (5) (+ve) numbers b/w 2 & 32 such that resulting sequence is an G.P.

Soln:



$a = 2$

$T_7 = ar^6 = 32$

$2r^6 = 32$

$r^6 = 16 = 2^4 \rightarrow r = \pm 2^{4/6} = \pm 2^{2/3}$

$r = 2^{\frac{2}{3}}$

$z = ar^3$

$= 2\left(2^{\frac{2}{3}}\right)^3$

$= 2 \times 4$

$z = 8$

MOST IMPORTANT OF GM's

Product of n-GM's inserted between 'a' & 'b' = $(ab)^{\frac{n}{2}} = (\sqrt{ab})^n$

INSERTING 'n' H.M.'s

Inserting 'n' H.M.'s between 'a' & 'b' :

$a, H_1, H_2, H_3 \ldots\ldots\ldots\ldots H_n, b$ → H.P.

Recip.

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \ldots\ldots \frac{1}{H_n}, \frac{1}{b}$ → A.P.

+d +d +d +d

first = $\frac{1}{a}$

last = $\frac{1}{b}$

Total terms = $n+2$.

$T_{n+2} = \frac{1}{a} + (n+1)d = \frac{1}{b}$

$(n+1)d = \frac{1}{b} - \frac{1}{a}$

Sum of Reciprocal of n-HM's = $\frac{1}{H_1} + \frac{1}{H_2} + \frac{1}{H_3} + \ldots\ldots + \frac{1}{H_n}$

$a, H_1, H_2, H_3 \ldots\ldots\ldots H_n, b \rightarrow$ H.P.

Recip.

$\frac{1}{a}, \frac{1}{H_1}, \frac{1}{H_2}, \frac{1}{H_3}, \ldots\ldots \frac{1}{H_n}, \frac{1}{b} \rightarrow$ A.P.

+d +d +d +d

first $= \frac{1}{a}$

last $= \frac{1}{b}$

Total terms $= n+2$.

$T_{n+2} = \frac{1}{a} + (n+1)d = \frac{1}{b}$

$(n+1)d = \frac{1}{b} - \frac{1}{a}$

Sum of Reciprocal of n-HM's $= \frac{1}{H_1} + \frac{1}{H_2} + \frac{1}{H_3} + \ldots\ldots + \frac{1}{H_n}$

$= \left(\frac{1}{a} + d\right) + \left(\frac{1}{a} + 2d\right) + \left(\frac{1}{a} + 3d\right) + \ldots + \left(\frac{1}{a} + nd\right)$

$= n\left(\frac{1}{a}\right) + d(1+2+\ldots+n)$

$= \frac{n}{a} + d(n+1)\frac{n}{2} = \frac{n}{a} + \left(\frac{1}{b} - \frac{1}{a}\right)\frac{n}{2} = \frac{n}{a} + \frac{n}{2b} - \frac{n}{2a} = \frac{n}{2a} + \frac{n}{2b}$

$= \frac{n}{2}\left(\frac{1}{a} + \frac{1}{b}\right)$

MOST IMPORTANT OF HM's

Sum of reciprocal of all HM's between 'a' & 'b' = $\frac{n}{2}\left(\frac{1}{a}+\frac{1}{b}\right)$

QUESTION PRACTICE

Question [JEE Mains-2023]

If n arithmetic means are inserted between a and 100 such that the ratio of the first mean to the last mean is $1:7$ and $a + n = 33$, then the value of n is



A 21

B 22

C 23

D 24

Question [JEE Mains-2021]

Let $A_1, A_2, A_3, \ldots\ldots$ be squares such that for each $n \geq 1$, the length of the side of A_n equals the length of diagonal of A_{n+1}. If the length of A_1 is 12cm, then smallest value of A_n for which area of A_n is less than one, is



DO IT BY YOURSELF (DIBY)

GEOMETRIC PROGRESSION

DIBY-34 (error!)

Let $a_1, a_2, a_3, \ldots$ be a G. P. of increasing positive numbers. If $a_3 a_5 = 729$ and $a_2 + a_4 = \frac{117}{4}$, then $24(a_1 + a_2 + a_3)$ is equal to [JEE Mains-2025]

(A) 131 (B) 130 (C) 129 (D) 128

[Ans. : (C)]

Check!

DIBY-35

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive terms. If $a_1 a_5 = 28$ and $a_2 + a_4 = 29$, then a_6 is equal to: [JEE Mains-2025]

(A) 812 (B) 526 (C) 784 (D) 628

[Ans. : (C)]

DIBY-36

Let a and b be two distinct positive real numbers. Let 11th term of a GP, whose first term is a and third term is b, is equal to p^{th} term of another GP, whose first term is a and fifth term is b. Then p is equal to [JEE Mains-2024]

(A) 20 (B) 25 (C) 21 (D) 24

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-37

Let $a_1, a_2, a_3, \ldots$ be a G.P. of increasing positive numbers. Let the sum of its 6th and 8th terms be 2 and the product of its 3rd and 5th terms be $\frac{1}{9}$.

Then $6(a_2 + a_4)(a_4 + a_6)$ is equal to [JEE Mains-2023]

(A) $2\sqrt{2}$ (B) 2 (C) $3\sqrt{3}$ (D) 3

[Ans. : (D)]

DIBY-38

In an increasing geometric progression of positive terms, the sum of the second and sixth terms is $\frac{70}{3}$ and the product of the third and fifth terms is 49. Then the sum of the 4th, 6th and 8th terms is: [JEE Mains-2024]

(A) 96 (B) 78 (C) 91 (D) 84

[Ans. : (C)]

GEOMETRIC PROGRESSION

DIBY-39

Find the four successive terms of a. G.P. of which the 2nd term is smaller than the first by 35 and the 3rd term is larger than the 4th by 560.

(A) $7, -28, 112, -448$

(B) $5, -20, 80, -320$

(C) $7, 28, 112, 448$

(D) $-\frac{35}{3}, -\frac{140}{3}, -\frac{560}{3}, -\frac{2240}{3}$

[Ans. : A, D]

DIBY-40

If a_1, a_2, a_3 are in G.P. such that $a_1 + a_2 + a_3 = 13$ and $a_1^2 + a_2^2 + a_3^2 = 91$. Find a and r.

(A) $(a = 1, r = 2)$ or $\left(a = 4, r = \frac{1}{2}\right)$

(B) $(a = 3, r = 1)$ or $\left(a = \frac{1}{3}, r = 9\right)$

(C) $(a = 2, r = 3)$ or $\left(a = 9, r = \frac{1}{2}\right)$

(D) $(a = 1, r = 3)$ or $\left(a = 9, r = \frac{1}{3}\right)$

[Ans. : (D)]

GEOMETRIC PROGRESSION

DIBY-41

If $S_1, S_2, S_3, \ldots . S_p$ are the sums of infinite G.P. whose first terms are $1, 2, 3, \ldots \ldots p$, and whose common ratios are $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots \ldots \frac{1}{p+1}$ respectively, then find the value of $S_1 + S_2 + S_3 + \cdots S_p$.

(A) $\frac{p(p+1)}{2}$ (B) $\frac{(p+1)(p+3)}{2}$ (C) $\frac{p}{2}(p+3)$ (D) $p(p+3)$

[Ans. : (C)]

DIBY-42

The sum of an infinite number of terms of a G.P. is 15 and the sum of their squares is 45. Find the series.

(A) $5, \frac{10}{3}, \frac{20}{9}, \frac{40}{27}, \ldots$ (B) $3, 2, \frac{4}{3}, \frac{8}{9}, \ldots$

(C) $5, \frac{5}{3}, \frac{5}{9}, \frac{5}{27}, \ldots$ (D) $3, \frac{6}{5}, \frac{12}{25}, \ldots$

[Ans. : (A)]

It is never about end result

It's always about Journey

Thank You

#futureIITians